Dynamic Programming

Part 3: Exercise 3 & 4 (week 16)

Mikkel Reich

University of Copenhagen, Department of Economics

Plan for today

- Solving the canonical Buffer-Stock model with EGM
  - Much more realistic Consumption-Saving model
  - Today: extension with life-cycle aspect
  - Often: Infinite horizon in macro (e.g. in HANK)
- Structural estimation of the Buffer-Stock model
  - Maximum Likelihood Estimation (MLE)
  - Method of Simulated Moments (MSM)

Exercise 3: Solving the Buffer-Stock model with EGM

Like extended version of Deaton's model.

$$V_t(M_t, P_t) = \max \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(M_{t+1}, P_{t+1}) \right] \right\}$$

$$\text{s.t.} \quad M_{t+1} = R(M_t - C_t) + \xi_{t+1} P_{t+1}$$

$$P_{t+1} = P_t GL_t \psi_{t+1}$$

$$A_t = M_t - C_t \geq P_t \max\{-\lambda_t, -\Omega_t\}$$

Deaton: only transitory income shocks ξ_{t+1} (like $P_t = 1 \forall t$).

+ "Permanent" income $\rightarrow$ Additional state variable P_t

+ Model realistic unemployment (next slide)

+ More realistic borrowing-constraint: $\Omega_t = 0$ for $t \geq T_R$

For $t < T_R$, Ω_t defined by the recursion $\Omega_t = \frac{\Omega_{t+1} + \text{min_income}_{t+1}}{R}$

Exercise 3: Income shocks and unemployment risk

Before retirement $t < T_R$:

Permanent shock: $\psi_{t+1} \sim \exp(\mathcal{N}(-0.5\sigma_\psi^2, \sigma_\psi^2)) \Rightarrow \mathbb{E}[\psi_{t+1}] = 1$

Transitory shock: $\xi_{t+1} = \begin{cases} \mu & \text{with prob. } \pi \\ \frac{\epsilon_{t+1} - \pi\mu}{1-\pi} & \text{with prob. } 1-\pi \end{cases}$

With $\epsilon_{t+1} \sim \exp(\mathcal{N}(-0.5\sigma_\xi^2, \sigma_\xi^2)) \quad \Rightarrow \mathbb{E}[\epsilon_{t+1}] = 1$

$\Rightarrow \mathbb{E}[\xi_{t+1}] = \pi\mu + (1-\pi)\frac{\mathbb{E}[\epsilon_{t+1}] - \pi\mu}{1-\pi} = 1$

After retirement $t \geq T_R$: No income risk $\Rightarrow \psi_{t+1} = \xi_{t+1} = 1$

Implementation: quadrature for ξ by "adding" node with $\xi = \mu$

$$\xi^{GH} = \left[\mu, \xi_1^{GH}, \ldots, \xi_n^{GH}\right], \quad w^{GH,\xi} = \left[\pi, (1-\pi)w_1^{GH,\xi}, \ldots\right]$$

Exercise 3: Gettin' rid of a state variable using homogeneity

Dynamic Programming $\rightarrow$ Curse of dimensionality

... Not a big problem here but useful exercise

We know $u(C_t) = \frac{C_t^{1-\rho}}{1-\rho}$ is homogeneous of degree $1-\rho$

$$u(\lambda C_t) = \frac{(\lambda C_t)^{1-\rho}}{1-\rho} = \lambda^{1-\rho}\frac{C_t^{1-\rho}}{1-\rho} = \lambda^{1-\rho}u(C_t)$$

State transition $M_{t+1}(\cdot), P_{t+1}(\cdot)$ are homogeneous of degree 1

$$M_{t+1}(\lambda M_t, \lambda C_t, \lambda P_{t+1}) = \lambda M_{t+1}(M_t, C_t, P_{t+1})$$

$$P_{t+1}(\lambda M_t, \lambda C_t, \lambda P_t) = \lambda P_{t+1}(M_t, C_t, P_t)$$

(even borrowing-constraint homogeneous of degree 1)

Gettin' rid of a state variable: Homogeneity of value function

Define aux. value and policy function(-ish):

$$v_t(M_t, P_t) = \lambda^{-(1-\rho)} V(\lambda M_t, \lambda P_t), \quad C_t = \lambda c_t$$

1) go back to $V(M_t, P_t)$ 2) multiply inputs by λ 3) insert $C_t = \lambda c_t$

$$V_t(\lambda M_t, \lambda P_t) = \max_{c_t} \left\{ \lambda^{1-\rho} \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(\lambda M_{t+1}, \lambda P_{t+1}) \right] \right\}$$

4) divide both sides with $\lambda^{1-\rho}$

$$\overbrace{\lambda^{-(1-\rho)} V_t(\lambda M_t, \lambda P_t)}^{=v_t(M_t, P_t)} = \max_{c_t} \left\{ \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[\overbrace{\lambda^{-(1-\rho)} V_{t+1}(\lambda M_{t+1}, \lambda P_{t+1})}^{v_{t+1}(M_{t+1}, P_{t+1})} \right] \right\}$$

Same Bellman equation $\rightarrow$ Same unique policy

$$\Rightarrow \quad \overbrace{v_t(M_t, P_t)}^{=\lambda^{-(1-\rho)} V_t(\lambda M_t, \lambda P_t)} = V_t(M_t, P_t) \quad \Leftrightarrow V_t(\lambda M_t, \lambda P_t)) = \lambda^{1-\rho} V_t(M_t, P_t)$$

$\rightarrow$ $V_t(\cdot, \cdot)$ is homogeneous of degree $1 - \rho$

Gettin' rid of a state variable: Bellman in "ratio-form"

Bottom line: Bellman "inherits" homogeneity from utility function

Normalize Bellman with $\frac{1}{P_{t+1}^{1-\rho}}$, def. $x_t = \frac{X_t}{P_t}$, $v_t(m_t) = V_t(M_t, 1)$

$$V_t\left(M_t, P_t\right) \frac{1}{P_{t+1}^{1-\rho}} = \max_{C_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(M_{t+1}, P_{t+1}) \right] \right\} \frac{1}{P_{t+1}^{1-\rho}}$$

$$V_t(M_t, P_t) \frac{1}{P_t^{1-\rho}} = \max_{C_t} \left\{ \frac{\left(\frac{C_t}{P_t}\right)^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[\left(GL_t \psi_{t+1}\right)^{1-\rho} V_{t+1}(M_{t+1}, P_{t+1}) \frac{1}{P_{t+1}^{1-\rho}} \right] \right\}$$

$$V_t\left(\frac{M_t}{P_t}, 1\right) = \max_{c_t} \left\{ \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[\left(GL_t \psi_{t+1}\right)^{1-\rho} V_{t+1}\left(\frac{M_{t+1}}{P_{t+1}}, 1\right) \right] \right\}$$

$$v_t(m_t) = \max_{c_t} \left\{ \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[\left(GL_t \psi_{t+1}\right)^{1-\rho} v_{t+1}(m_{t+1}) \right] \right\}$$

Same with state-transition and borrowing-constraint:

$$m_{t+1} = \frac{1}{GL_t \psi_{t+1}} R \overbrace{a_t}^{=m_t - c_t} + \xi_{t+1}, \qquad a_t \geq \max\{-\lambda_t, -\Omega_t\}$$

Exercise 4: Estimating the Buffer-Stock model

Framework is the same as Harold Zurcher:

Outer loop: Find parameters that makes model fit data

Inner loop: Solve model given parameter guess

Exercise 3: Can solve Buffer-Stock model fast given parameters.

Outer loop: Use your favorite tools from econometrics

- MLE: Impose distribution on e.g. "measurement errors": for some reason observed outcomes deviates from optimal policy
- GMM: Impose that certain moments equate

Today: We work with simulated data..

PS: Estimation = Calibration + Standard Errors

Exercise 4: Maximum Likelihood Estimation

Idea: Do MLE on consumption for cross-section in $t = 20$

Consider normal measurement errors in logs for $t = 20$:

$$\epsilon_{i,20}(\theta) = \ln(C_{i,20}) - \ln(C^*_{i,20}(M_{i,20}, P_{i,20}|\theta)) \sim \mathcal{N}(0, \sigma_\epsilon)$$

Simulation (using θ_0): obs. for $t = 20$: $\{C_{i,20}, M_{i,20}, P_{i,20}\}_{i=1}^{N_{sim}}$

Solving model: policy function $c^*_{20}(m_{20}|\theta)$

1. Observe (a.o.) $m_{i,20} = \frac{M_{i,20}}{P_{i,20}}$ and $C_{i,20}$
2. Call policy on state: $c^*_{20}(m_{i,20}|\theta)$ by interpolating
3. Transform policy: $C^*_{20}(m_{i,20}|\theta) = c^*_{20}(m_{i,20}|\theta)P_{i,20}$
4. Compute error $\epsilon_{i,20}(\theta) = \ln(C_{i,20}) - \ln(C^*_{20}(m_{i,20}|\theta))$ for obs i

Minimize neg. log-likelihood: $-\frac{1}{2}\ln(2\pi\sigma^2_\epsilon)N_{sim} - \sum_{i=1}^{N_{sim}} \frac{\epsilon^2_{i,20}(\theta)}{2\sigma^2_\epsilon}$

Exercise 4: Method of Simulated Moments

Other idea: match average savings moment for ages: $40 - 55$

Model "begins" for 25yr olds $\rightarrow$ Consider $t \in \{16, \ldots, 31\}$

"Data": obs. $\{C_{i,t}, M_{i,t}, P_{i,t}\}_{i=1,\ldots,N_{sim},t=16,\ldots,31}$ (from sim. w. θ_0)

$\rightarrow$ Compute "true" moments of data: $\Lambda_t^d = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} A_{i,t}$

Solve model + Simulate given parameters θ

$\rightarrow$ Compute moments: $\Lambda_t^m(\theta) = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} M_{i,t}(\theta) - C_{i,t}^*(M_{i,t}|\theta)$

(sometimes do multiple simulations and average)

Solve "minimum-distance"/"moment error" problem:

$$\hat{\theta} = \arg\min_{\theta} (\Lambda_t^d - \Lambda_t^m(\theta))' W (\Lambda_t^d - \Lambda_t^m(\theta))$$

Your time to shine!